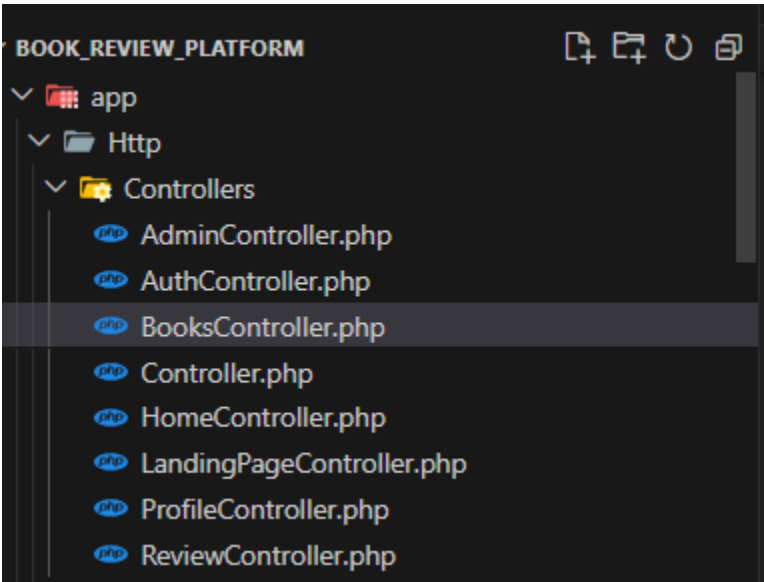
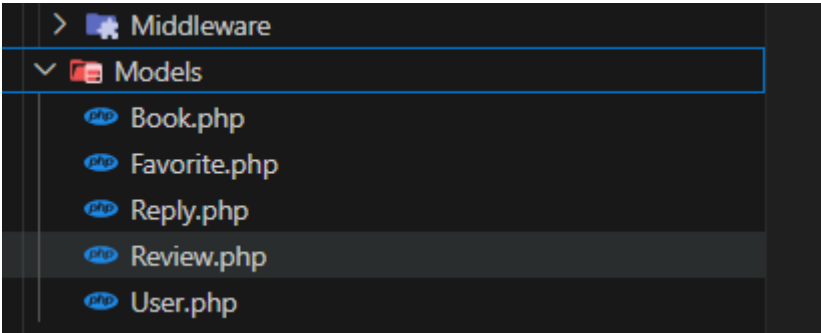


I. Documentation

- Controllers in our project



- The models in our project



- I will only discuss some of the controllers, First is the home controller, the HomeController serves as the intermediary between the application's logic and the Book model, facilitating data retrieval for the home view. It uses the Book model to fetch two specific sets of data: the top 4 books with the highest average ratings from the related reviews (review model/ table) using the withAvg method, sorted in descending order of their average ratings (popularNow), and the 4 most recently released books, determined by sorting the release_date field in descending order (latestBooks). These collections are then passed to the home view using the compact function, enabling the view to dynamically display this information to the user.
- I will no longer discuss the logic in the view/blade file (the foreach that loops to the returned associative array—through the compact function)
- Because of the models its much easier to make methods/functions in retrieving than manually using raw SQL code, as long as the models have relation functions in to them which is seen in the snapshot I provided below

book-details.blade.phpweb.phpBooksController.phpHomeController.php X

app > Http > Controllers > HomeController.php > ...

```
1  <?php
2
3  namespace App\Http\Controllers;
4
5  use App\Http\Controllers\BooksController;
6  use Illuminate\Http\Request;
7  use Illuminate\Support\Facades\DB;
8  use App\Models\Book;
9
10
11  2 references | 0 implementations
12  class HomeController extends Controller
13  {
14      1 reference | 0 overrides
15      public function index(): View
16      {
17          $popularNow = Book::withAvg(relation: 'reviews', column: 'rating') // Calculate the average rating from the 'reviews' relationship
18          ->orderByDesc(column: 'reviews_avg_rating') // Sort by the average rating in descending order
19          ->take(value: 4) // Limit to 4 books
20          ->get();
21
22          $latestBooks = Book::withAvg(relation: 'reviews', column: 'rating') // Calculate the average rating from the 'reviews' relationship
23          ->orderBy(column: 'release_date', direction: 'desc') // Order by release_date (newest first)
24          ->take(value: 4) // Get the 4 most recent books
25          ->get();
26
27          return view('home', data: compact(var_name: 'popularNow', var_names: 'latestBooks'));
28      }
29  }
```

book-details.blade.phpweb.phpBooksController.phpBook.php X

app > Models > Book.php > Book

```
3  namespace App\Models;
4
5  use Illuminate\Database\Eloquent\Model;
6  use Illuminate\Database\Eloquent\Factories\HasFactory;
7
8  41 references | 0 implementations
9  class Book extends Model
10  {
11      use HasFactory;
12
13      0 references
14      protected $table = 'books';
15
16      0 references
17      public $primaryKey = 'bookID';
18
19      0 references
20      public $timestamps = true;
21
22      0 references
23      protected $fillable = ['title', 'author', 'genre', 'description', 'cover'];
24
25      // Define the one-to-many relationship with Review
26      0 references | 0 overrides
27      public function reviews(): HasMany
28      {
29          return $this->hasMany(related: Review::class, foreignKey: 'bookID'); // The 'bookID' is the foreign key in the 'reviews' table
30      }
31  }
```

- Route assignment applied to the home view with the home controller

```
Route::get(uri: '/home', action: [HomeController::class, 'index'])->name(name: 'home.index');
```

• Rendered Page

It Elec 1 - Web Development: P

127.0.0.1:8000/home

127.0.0.1:8000/home

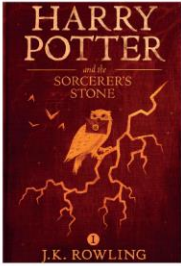
☆

Top Rated & Popular Now

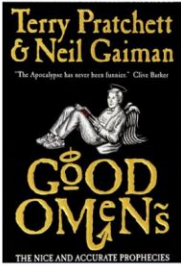
View All



Dungeon Crawler Carl
Matt Dinniman
5.0 ★★★★★



Harry Potter and the Sorcerer's Stone
J.K. Rowling
5.0 ★★★★★



Good Omens
Neil Gaiman and Terry Pratchett
5.0 ★★★★★



Mistborn: The Final Empire
Brandon Sanderson
5.0 ★★★★★

It Elec 1 - Web Development: P

127.0.0.1:8000/browse/books/

127.0.0.1:8000/browse/books/view-all/rating/popular

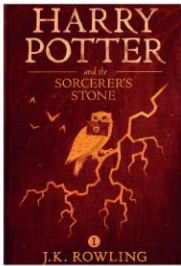
☆

Home / Browse Books / Books by Rating

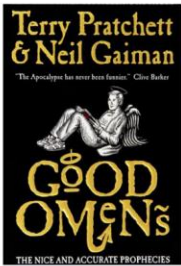
Top Rated & Popular Now



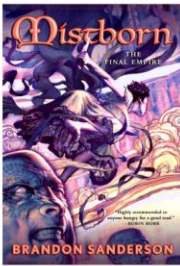
Dungeon Crawler Carl
Matt Dinniman
5.0 ★★★★★



Harry Potter and the Sorcerer's Stone
J.K. Rowling
5.0 ★★★★★



Good Omens
Neil Gaiman and Terry Pratchett
5.0 ★★★★★



Mistborn: The Final Empire
Brandon Sanderson
5.0 ★★★★★

It Elec 1 - Web Development: P

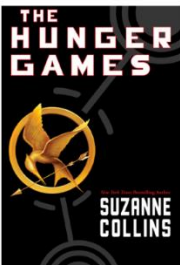
127.0.0.1:8000/home

127.0.0.1:8000/home

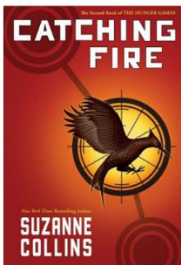
☆

Latest Books

View All



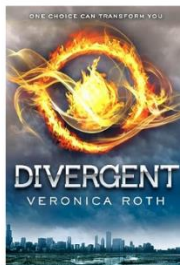
The Hunger Games
Suzanne Collins
0.0 ☆☆☆☆☆



Catching Fire
Suzanne Collins
4.3 ★★★★★☆

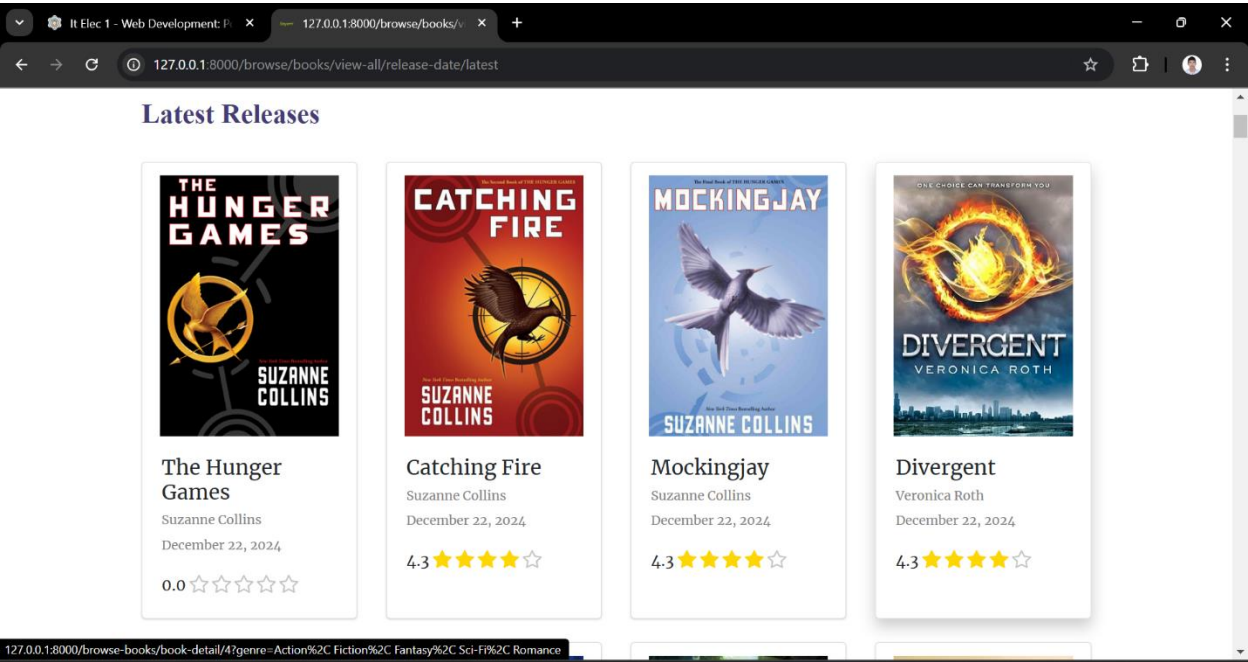


Mockingjay
Suzanne Collins
4.3 ★★★★★☆



Divergent
Veronica Roth
4.3 ★★★★★☆

127.0.0.1:8000/browse-books/book-detail/4?genre=Action%2C Fiction%2C Fantasy%2C Sci-Fi%2C Romance

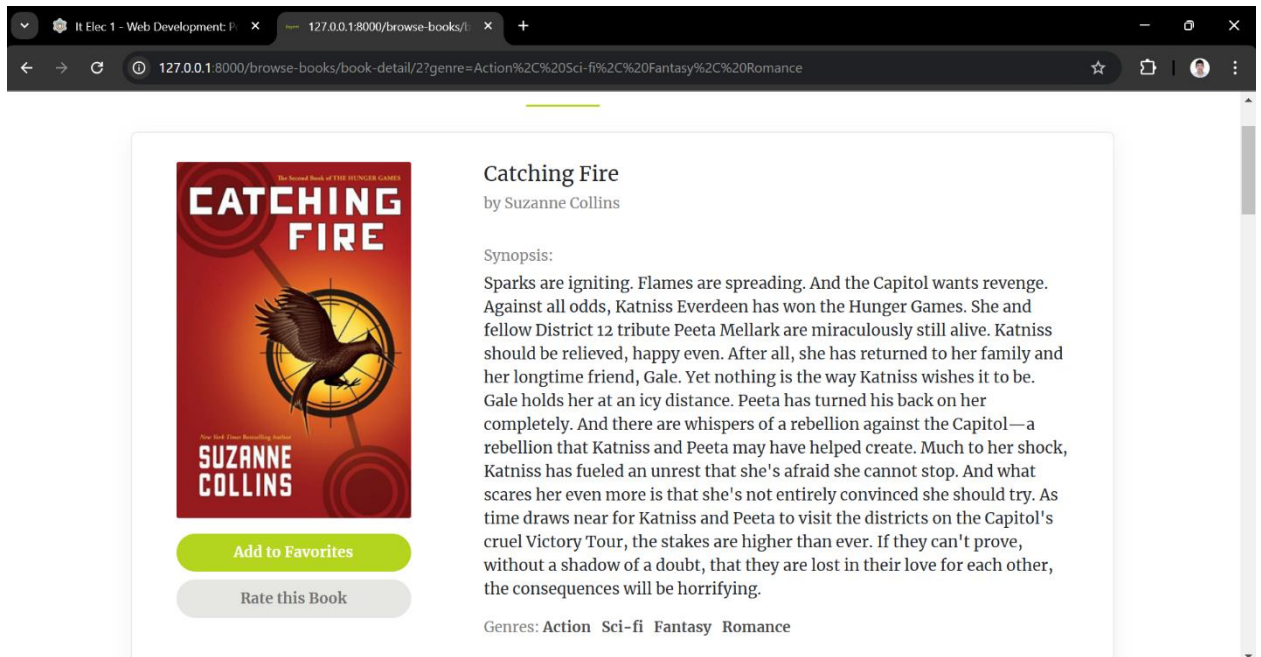


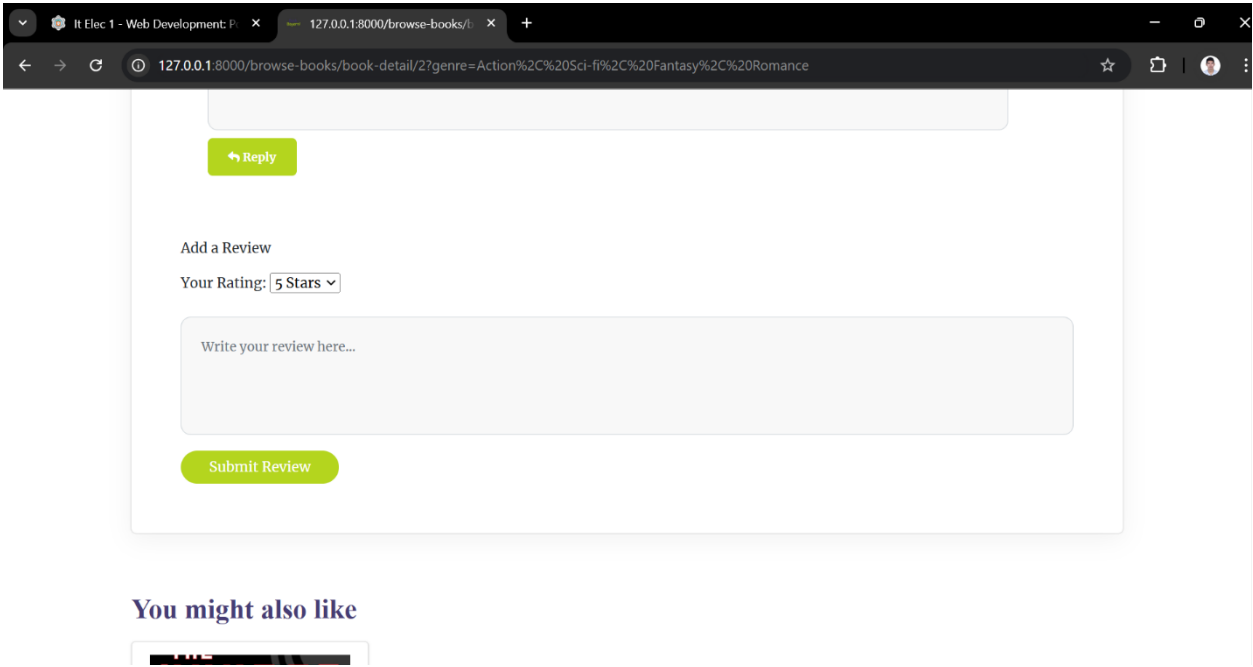
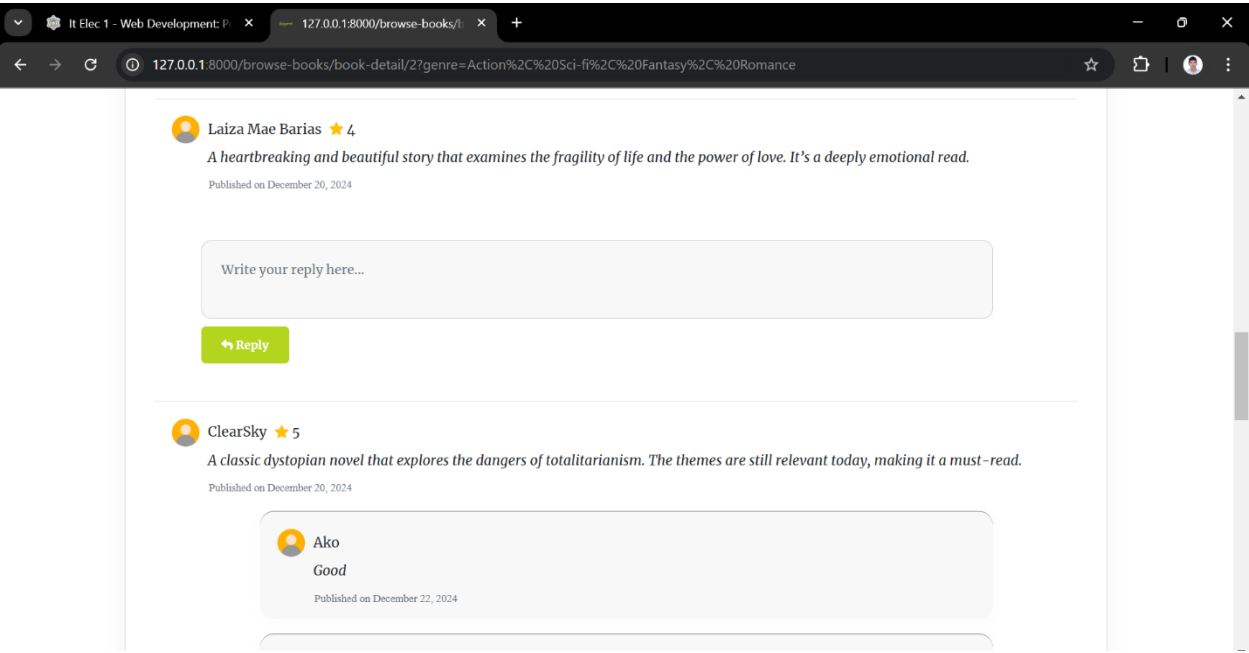
- Another controller I will discuss is the books controller and the review controller (since in a certain view—bookDetails I used/ applied both the controller).
- The showBookDetail method in the BookController is responsible for displaying detailed information about a specific book by integrating data from various sources (can be considered na as a composite since it integrates me other methods to return multiple var/collection to the same view). It retrieves the book's main details by calling the show method within the same controller, fetches the associated ratings and reviews by instantiating and using the showRatings method from the ReviewController, and generates a list of recommended books related to the given book. Finally, it combines all this data—book details, ratings, the book's ID, and the recommended books—into a single array using array_merge, and passes it to the books.book-details view for rendering.
- Because of that single method, the route definition is much simplier and in only a single line (na amaze din ako, pa iba iba pa ako ng uri para mag work pede din pala yun)

```
book-details.blade.php | web.php | BooksController.php | Book.php
app > Http > Controllers > BooksController.php > BooksController
12 class BooksController extends Controller
93 public function showByRating(): View
96     ->orderByDesc(column: 'reviews_avg_rating') // Order by rating
97     ->get();
98
99     return view(view: 'books.show-books-by-rating', data: compact(var_name: 'booksByRating'));
100 }
101
102
103
1 reference|0 overrides
104 public function showBookDetail($id): View
105 {
106     // Retrieve book details from BooksController
107     $bookDetails = $this->show(id: $id);
108
109     // Retrieve reviews from ReviewController
110     $reviewController = new ReviewController();
111     $ratings = $reviewController->showRatings(id: $id);
112
113     // Retrieve recommended books
114     $recommendedBooks = $this->recommendBooks(bookId: $id)->take(limit: 4)->toArray(); // Convert to array and limit to 4 books
115
116
117     // Return the view with the necessary data
118     return view(view: 'books.book-details', data: array_merge(
119         arrays: $bookDetails->getData(), // Pass book details as array
120         $ratings->getData(), // Pass ratings as array
121         ['id' => $id], // Pass the book id
122         ['recommendedBooks' => $recommendedBooks] // Pass recommended books as array
123     ));
124 }
125
126
127

Route::get(uri: '/browse-books/book-detail/{id}', action: [BooksController::class, 'showBookDetail'])->name(name: 'books.bookDetail');
```

- Rendered Page





- that's all po, hirap ng review and reply feature sir ha HAHAHAHAHAHHA